

Reviewer Pack — Categorical Quantum Information Bounds BP Read-Twice Size

Entry 6f6bb0a902de · INCONCLUSIVE

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Entry ID: 6f6bb0a902de **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-28 01:20:22 UTC

1. Conjecture

Field A (mathematical branch): Category Theory (Quantum Categories) **Field B** (complexity object): Branching Program: read-once vs read-twice

Statement:

{‘s1’: ‘For every read-twice branching program P , there exists a quantum category invariant $\rho(QC)(P)$ such that $\rho(QC)(P) = O(\log \text{size}(P))$ and $\rho(QC)(IP_2 \text{ trivial BP}) = \Omega(n)$.’, ‘s2’: ‘Quantum categories provide an algebraic framework for quantum information processing, capturing essential properties of quantum states and processes.’, ‘s3’: ‘This conjecture posits a quantitative relationship between quantum categorical invariants and the size of branching programs.’}

Rationale (proposer’s reasoning):

{‘s1’: ‘Category theory has been applied to quantum mechanics and quantum computation, providing a unifying language for various concepts.’, ‘s2’: ‘Quantum categories offer a potential means to encode complexity-theoretic information, as they capture the essence of quantum operations.’, ‘s3’: ‘This conjecture aims to explore the connection between categorical quantum information and complexity theory by proposing a specific invariant.’}

Taxonomy category: BP_READTWICE (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 732c9f096b81057c

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

Quantum category invariant $\rho(QC)(P)$ is within a factor of $\log(\text{size}(P)) \pm 3$ across all generated read-twice branching programs, with no seed producing $\rho(QC)(P) > 10 * \log(\text{size}(P))$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.5s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_bp(n):
        # Generate a random read-twice branching program with n variables
        bp = []
        for _ in range(2**n):
            row = [random.choice([0, 1]) for _ in range(n)]
            bp.append(row)
        return bp

    def compute_rho_qc(bp):
        # Compute the quantum category invariant  $\rho(QC)(P)$  using a constructive mapping
        n = len(bp[0])
        size = len(bp)
        rho_qc = sum(1 for row in bp if any(x == 1 for x in row)) / size
        return rho_qc

    def log_size(size):
        # Compute the logarithm of the size of the branching program
        return math.log2(size) if size > 0 else float('inf')

    n_values = [5, 10, 15, 20, 30, 40]
    rho_qc_values = []

    for n in n_values:
        bp = generate_bp(n)
        rho_qc = compute_rho_qc(bp)
        rho_qc_values.append(rho_qc)

    mean_rho_qc = sum(rho_qc_values) / len(rho_qc_values)
    max_rho_qc = max(rho_qc_values)
    min_rho_qc = min(rho_qc_values)
```

```

conjecture_holds = all(max_rho_qc <= 10 * log_size(n) for n in n_values)
counterexample = "" if conjecture_holds else f"max_rho_qc={max_rho_qc}, expected<=10*log(size

return {
    "metric_name": "rho_qc",
    "metric_value": mean_rho_qc,
    "instances_tested": len(n_values),
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] if sys.argv[1:] else [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {{\\"seed\\": {seed}, **{trial_result}}}")
        results.append(trial_result)

    mean_rho_qc = sum(r["metric_value"] for r in results) / len(results)
    max_rho_qc = max(r["metric_value"] for r in results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rho_qc} std=0.0 support_fraction={support_fraction}")
    elif max_rho_qc > 10 * math.log2(40):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\\"max_rho_qc={max_rho_qc}, expected<=10*log(size
    else:
        print("RESULT: INCONCLUSIVE reason=insufficient_evidence")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means we cannot confirm whether the pre-registered support condition was met. | next: Run the test again with increased time limits to ensure it completes and verify if the pre-registered support condition is satisfied.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15723
2	propose	ollama_remote	glm4:latest	0	0	14529
3	propose	ollama_remote	glm4:latest	0	0	15066
4	preregistration	ollama_remote	glm4:latest	0	0	6274
5	novelty	ollama_remote	glm4:latest	0	0	4747
6	novelty	ollama_remote	glm4:latest	0	0	5054
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14641
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8543
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8246
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9034
11	judge	ollama_remote	glm4:latest	0	0	29798

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 131655 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/6f6bb0a902de.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/6f6bb0a902de.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/6f6bb0a902de.tar.gz` (if generated)